

IB AAHL MATHEMATICS

INTEGRATION BY PARTIAL FRACTIONS

Set of 10 Multiple-Choice Questions

Instructions: Select the one correct answer for each question. Give exact answers where appropriate. Calculators are not required.

1. Which is the correct partial fraction decomposition of $(3x + 5)/(x^2 - 1)$?

- A. $2/(x - 1) + 1/(x + 1)$
- B. $4/(x - 1) - 1/(x + 1)$
- C. $1/(x - 1) + 4/(x + 1)$
- D. $3/(x - 1) + 2/(x + 1)$

2. Evaluate the integral: integral of $1/(x^2 - 1)$ dx.

- A. $(1/2) \ln|x - 1| + (1/2) \ln|x + 1| + C$
- B. $(1/2) \ln|(x - 1)/(x + 1)| + C$
- C. $\ln|x^2 - 1| + C$
- D. $\ln|x - 1| - \ln|x + 1| + C$

3. The expression $5/[x(x + 2)]$ is decomposed as $A/x + B/(x + 2)$. What is the value of A?

- A. $-5/2$
- B. -2
- C. $5/2$
- D. 5

4. Evaluate the integral: integral of $5/[x(x + 2)]$ dx.

- A. $(5/2)\ln|x| - (5/2)\ln|x + 2| + C$
- B. $(5/2)\ln|x| + (5/2)\ln|x + 2| + C$
- C. $5\ln|x| - 5\ln|x + 2| + C$
- D. $(5/2)\ln|x + 2| - (5/2)\ln|x| + C$

5. Which decomposition is appropriate for $(2x + 1)/[x(x - 1)^2]$?

- A. $A/x + B/(x - 1)$
- B. $A/x + B/(x - 1) + C/(x - 1)^2$
- C. $A/x + B/x^2 + C/(x - 1)$
- D. $A/[x(x - 1)] + B/(x - 1)^2$

6. Find the value of A if $3/[x(x - 1)] = A/x + B/(x - 1)$.

- A. -3
- B. -1
- C. 1
- D. 3

7. Evaluate the integral: integral of $1/[x(x - 1)] dx$.

- A. $\ln|x| + \ln|x - 1| + C$
- B. $-\ln|x| + \ln|x - 1| + C$
- C. $\ln|x| - \ln|x - 1| + C$
- D. $1/x - 1/(x - 1) + C$

8. Which statement about partial fractions is correct?

- A. Every rational function can be integrated directly without decomposition.
- B. A repeated linear factor requires a separate term for each power of the factor.
- C. A quadratic factor must always be split into two real linear factors.
- D. Partial fractions can only be used when the numerator has lower degree than the denominator.

9. Evaluate the integral: integral of $(x + 2)/(x^2 - 4) dx$.

- A. $\ln|x - 2| + C$
- B. $(1/2)\ln|x^2 - 4| + C$
- C. $\ln|x + 2| + C$
- D. $(1/2)\ln|(x - 2)/(x + 2)| + C$

10. For the integral of $(x^2 + 1)/[x(x - 1)] dx$, which first step is most appropriate?

- A. Apply integration by parts immediately.
- B. Factor the denominator and use polynomial division before partial fractions.
- C. Substitute $x = 1$.
- D. Take the square root of the denominator.

ANSWER KEY

Answers and brief checks

Question	Answer	Check / Key idea
1	B	$4/(x - 1) - 1/(x + 1)$
2	B	$(1/2)\ln (x - 1)/(x + 1) + C$
3	C	$A = 5/2$
4	A	$(5/2)\ln x - (5/2)\ln x + 2 + C$
5	B	Repeated factor $(x - 1)^2$ needs both powers.
6	A	$A = -3$
7	C	$1/[x(x - 1)] = -1/x + 1/(x - 1)$
8	B	Each power of a repeated linear factor gets its own term.
9	C	The integrand simplifies to $1/(x - 2)$.
10	B	The numerator degree is not lower than the denominator degree.

Topic coverage: decomposition with distinct linear factors, repeated factors, logarithmic integration, and the need for polynomial division.